

Hiding in text/plain sight: Security defences of Tor Onion Services

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Abstract

Tor Onion Services are a way to host websites and other internet services anonymously. Onion Services are often used to bypass internet censorship and provide information services to users in oppressive regimes. This paper presents an analysis of the security defences deployed on these Onion Services. Onion Services tend to have better security policy than sites on the clear web. However they lag behind in the deployment of HTTPS, a key defence to ensuring the security of users of such services.

1 Introduction

Tor (alternatively The Onion Router) is a decentralised network of hosts allowing communication across the internet in which neither the sender nor the receiver know anything about each other. It achieves this by encapsulating data in layers of encryption (the onion concept), and transiting data over at least 3 nodes. Each node is aware of its neighbours, such as the data's source or final destination, but no node is ever aware of both the source and destination of the data[1].

The Tor network supports so-called Onion Services. These act in most ways like normal websites, but provide much stronger cryptographic guarantees about privacy and data integrity than traditional internet websites. When an Onion Service is configured correctly it is near impossible to track who connects to it, and where the server hosting the site is[2].

Onion Services provide a vital way to circumvent internet censorship in repressive regimes and get accurate news into countries that lack freedom of the press[3]. Many newsrooms have a presence on the Onion Services network, including the BBC, Deutsche Welle, The Guardian, and ProPublica[4]–[7]. The anonymity of Onion Services also lends itself well to less ethical activities[8], however this should not draw away from its benefits to society.

Given the danger users can put themselves in by accessing content banned in their country[9], it is imperative that these users stay as anonymous as possible. It is also important that operators of Onion Services remain as anonymous as possible, less they are targeted by regimes unfriendly to them[10]. This research aims to analyse various security and privacy methods employed by Onion Services, their prevalence across the Onion Service network, and categorise which types of sites tend to employ which methods.

2 Research question

This paper aims to analyse the security defences deployed on Onion Services, and compare them to the state of security defences on sites on the clear web. The hypothesis considered in this paper are as follows:

1. Onion Services are as likely to use HTTPS as other sites.
2. Single Onion Services are as likely to use HTTPS as other Hidden Services.
3. Onion Services use the same versions of TLS as other sites.
4. Onion Services use the certificate key lengths as other sites.
5. Onion Services are as likely to use Extended Validation certificates as other sites.
6. Onion Services are as likely to use Strict Transport Security as other sites.
7. Onion Services are as likely to use a Content Security Policy as other sites.

8. Onion Services are as likely to use a Permissions Policy as other sites.
9. Onion Services are as likely to use X-Frame-Options as other sites.
10. Onion Services are as likely to use X-Content-Type-Options as other sites.
11. Onion Services are as likely to use a Referrer Policy as other sites.

2.1 Scope

Onion Services can host a multitude of services capable of running over TCP, not just websites using HTTP(S)[11]. In order to limit the scope of work only HTTP services on the Tor network will be analysed. The class of security and privacy problems that can apply to internet services other than HTTP are distinct and out of scope for this paper.

3 Related work

3.1 OnionScan

OnionScan¹ was a project that enumerated all Onion Services hosting websites and scanned for security vulnerabilities. Since the deprecation of version 2 Onion Services in 2020[12] it's no longer possible to enumerate all of the Onion Services on the Tor network, due to the use of a blinded public key for the .onion address generation[11].

The analysis completed by OnionScan provides an insight into what data would be useful to collect about Onion Services to analyse their security, and methods to locate Onion Services to analyse[13].

3.2 Qualys SSL Labs

Qualys SSL Labs² is a tool for analysing the correctness of a site's implementation of TLS for HTTPS. Whilst there may appear to be no security benefit of using HTTPS over Tor, as connections to Onion Services are already end-to-end encrypted[11], this is not the case[14]. Of particular note is that the Tor tunnel may terminate before the HTTP reaches the final web server (as in the case of sites using the Enterprise Onion Toolkit³), and HTTPS provides security guarantees about this final step in accessing the site. A user may also not connect directly to the Tor network when accessing a Onion Service but instead use a Tor proxy somewhere on their local network via SOCKS5[15]; HTTPS secures this part of the connection as well. Qualys SSL labs tests a website for various possible misconfiguration of its TLS setup that could impact privacy or security of the site or its users[16].

Browsers also behave very differently for HTTP and HTTPS sites. Features such as microphone and webcam access, WebUSB, and geo-location (to name but a few) are only available on HTTPS enabled sites[17]. While the Tor Browser may make patches to enable such features on non-HTTPS Onion Services, the Tor Project team is a lot smaller than the teams behind the likes of Mozilla Firefox and Google Chrome. Using the codepath that's been developed by the vastly more mature team with stricter code review flows will likely result in better security overall.

Whilst SSL Labs is incapable of scanning Onion Services, their reports on the state of TLS on the clear web provide insight into what misconfigurations should be investigated in Onion Services.

3.3 Security Headers

Servers can send various HTTP headers back to the client to enforce security policies on their site, to aid in preventing man-in-the-middle, code injection, etc. TLS protects the contents of a data exchange, but does not entirely influence the security policy applied by a web browser[18]. Security Headers⁴ is a tool developed by Scott Helme that grades sites on their use of various security policy headers.

Security Headers is incapable of scanning Onion Services, but provides insight into useful metrics to analyse, and datasets from it can be used as a comparator for security between Onion Services and sites on the clear web.

¹<https://onionscan.org>

²<https://www.ssllabs.com/ssltest/>

³<https://github.com/alecmuffett/eotk>

⁴<https://securityheaders.com>

3.4 Discovering Onion Services

Onion Services are - by their very nature - hidden. This makes collecting a list of Onion Services to analyse a somewhat difficult prospect. It would be impossible to know of every Onion Service in existence. Onion Service owners do however have to publicise their Onion Services URL somewhere, or nobody would ever connect to it. Owners often publish their Onion Service URLs on sites such as The Hidden Wiki⁵, or submit them to the Onion Search Engine⁶[19]. These collections can be used to gather a list of sites to analyse.

Certificate Transparency logs can also be used as a source to discover Onion Services[20]. For any website to obtain a publicly trusted TLS certificate the certificate must be submitted to a public and searchable log of certificates.

4 Methodology

This steps required to collect data for this research can be broken down into the following broad stages:

1. collation of a list of Onion Services
2. collection of data about each Onion Service discovered
3. collection of the same data about clear web sites

4.1 Collecting Onion Services

A web crawler was developed that searches through Onion Services for links to other Onion Services, akin to how Googlebot[21] - and other search engines - crawl and discover content on the public internet. Given the slow speed of connections over the Tor network[22], and the long time it takes to initiate connections to new Onion Service; the collection was run in a highly parallelised manner with 50 Tor clients and 50 crawlers balancing load between the Tor clients. This was all coordinated by message passing over RabbitMQ. The source code for this crawler is available on GitHub⁷.

Certificate Transparency logs were also searched also be searched for any certificates issued to Onion Services. Certificate Transparency is a log consisting of every publicly trusted X.509 certificate ever issued[23], and thus included every Onion Service with a publicly trusted certificate. To complete this search Sectigo's crt.sh⁸ was used, a database of every certificate ever logged in any widely trusted Certificate Transparency log.

4.1.1 Limitations

This methodology of discovering Onion Services does have its limitations. Firstly the web crawling approach only discovers sites that people actively publicise on other sites. This is true of the clear web too, however Onion Services wish to stay entirely hidden, as would be expected from sites using such facilities as Tor. This likely had an effect on the type of sites discovered, and the number.

Discovering sites via Certificate Transparency likely resulted in an over-count of the prevalence of HTTPS on Onion Services. The dataset used for analysis will contain every single Onion Service with HTTPS, but not contain every single one without.

4.2 Collection of data about Onion Services

The following data was collected about each Onion Service in the discovered dataset.

1. which ports are open
 - (a) HTTP (port 80)
 - (b) HTTPS (port 443)
 - (c) SSH (port 22)
 - (d) FTP (port 21)
 - (e) SMTP (port 25)
2. if HTTPS is offered

⁵<https://thehiddenwiki.org>, further context <https://web.archive.org/web/20150628213012/https://www.deepdotweb.com/2014/11/15/the-hidden-wiki-seized/>

⁶<https://onionsearchengine.com>

⁷<https://github.com/TheEnbyperor/shadow-weaver>

⁸<https://crt.sh>

- (a) which versions of TLS are supported
 - (b) supported TLS cipher suites
 - (c) downgrade attack prevention[24]
 - (d) support for TLS compression
 - (e) support for TLS 1.3 early data
 - (f) is the service vulnerable to Heartbleed
 - (g) is the service vulnerable to CCS injection
 - (h) is the service vulnerable to Client Renegotiation DOS
 - (i) is the service vulnerable to ROBOT
 - (j) is the X.509 certificate presented an extended validation certificate
 - (k) is the X.509 certificate presented a publicly trusted certificate
3. is the service a Single Onion Service⁹
 4. Strict Transport Security[25]
 5. Content Security Policy[26]
 6. Permissions-Policy[27]
 7. X-Frame-Options
 8. X-Content-Type-Options
 9. Referrer Policy[28]
 10. Cross Origin Embedder Policy[29]
 11. Cross Origin Opener Policy[29]
 12. Cross Origin Resource Policy[29]
 13. Onion-Location¹⁰
 14. Apache mod_status¹¹ [13]

A tool called OnionSec was developed to complete this analysis. The source code for this tool is available on GitHub¹². The tool is also available as a service on the web for anyone wishing to run the same set of tests against their Onion Service and receive a report on its security¹³.

4.3 Collection of data about clear web sites

For comparison against more standard clear web sites data collected by Scott Helme in his Crawler.Ninja¹⁴ were used. This dataset contains security metrics of the top 1 million sites on the internet, as ranked by Amazon's Alexa.

5 Results

The data collected about Onion Services is available on GitHub¹⁵. Throughout this paper a confidence of 5% is used to reject H_0 .

Of the 957 Onion Services discovered, 309 (32.29%) could not be scanned for one reason or another, such as the site being offline, requiring client authentication, or taking too long to respond (over 30 seconds). Of those scanned the open ports are displayed in table 1. The

Apache mod_status The proportion of Onion Services with exposed Apache mod_status pages is shown in table reftable:mod-status. This is comparable to the 7-10% detected by OnionScan in 2017[30], showing no improvement, and perhaps even a worsening on this front. No detailed data was released by OnionScan so a statistical test of this unfortunately cannot be conducted.

⁹A Single Onion Service is an Onion Service that is configured for services that do not require anonymity themselves, but want it for clients connecting to their service. Single Onion Services use only three hops in the circuit rather than the typical six hops for onion services.

¹⁰<https://community.torproject.org/onion-services/advanced/onion-location/>

¹¹https://httpd.apache.org/docs/2.4/mod/mod_status.html

¹²<https://github.com/theenbyperor/onion-sec>

¹³<https://onionsec.net>

¹⁴<https://crawler.ninja>

¹⁵<https://gist.github.com/TheEnbyperor/90ba14517a0a8d184f8744252c6a6e8e>

Port	Number of sites	Proportion of sites scanned
HTTP (80)	614	94.75%
HTTPS (443)	198	30.56%
SSH (22)	4	0.62%
FTP (21)	0	0.00%
SMTP (25)	4	0.62%

Table 1: Open ports on Onion Services

Port	Number of sites	Proportion of sites scanned
mod_status Server Info	61	9.41%
mod_status Server Status	69	10.65%

Table 2: Apache mod_status on Onion Services

5.1 HTTPS support

Hypothesis 1 Onion Services are as likely to use HTTPS as other sites. The proportion of sites supporting HTTPS are shown in 3. The χ^2 contingent test between Onion services and other sites has a p-value of < 0.05 (8.16×10^{-225}), thus rejecting the null hypothesis and confirming that Onion Services are less likely to use HTTPS than sites on the clear web.

HTTPS supported	Type of site	Number	Proportion of sites scanned
Yes	Onion Service	198	30.56%
No	Onion Service	450	69.44%
Yes	Clear web	804,999	80.50%
No	Clear web	195,001	19.50%

Table 3: HTTPS support on different sites

Hypothesis 2 Single Onion Services are as likely to use HTTPS as other Onion Services. The proportion of sites being a Single Onion Service is shown in table 4. The χ^2 contingent test between Single Onion Services and HTTPS has a p-value of < 0.05 (0.00407), thus rejecting the null hypothesis and confirming that Single Onion Sites are more likely to use HTTPS than other Onion Services.

Single Onion status	Number of sites	Proportion of sites scanned
Not a Single Onion Service	504	77.78%
Single Onion Service	144	22.22%
Single Onion Service with HTTPS	124	19.14%
Single Onion Service without HTTPS	20	3.09%
Not a Single Onion Service with HTTPS	74	11.42%
Not a Single Onion Service without HTTPS	430	66.36%

Table 4: Single Onion Status of Onion Services

Hypothesis 3 Onion Services use the same versions of TLS as other sites. Of the Hidden Services supporting HTTPS the proportions of them supporting different TLS versions are shown in table 5. The same metrics for clear web sites are shown in table 6. The χ^2 contingent test between Onion Services and the clear web has a p-value of < 0.05 (6.42×10^{-35}), thus rejecting the null hypothesis and confirming that Onion Services use different versions of TLS than on the clear web. Onion Services tend to use TLS 1.2 whilst those on the clear web have a clear preference for TLS 1.3.

Hypothesis 4 Onion Services use the certificate key lengths as other sites. Of the Onion Services supporting HTTPS the proportions of them having different key lengths are shown in table 7. The same metrics for clear web sites are shown in table 8. Only RSA public keys are considered, those not counted in the tables used Elliptic Curve or another kind of key, for which key length has less of an effect on security. The χ^2 contingent test between Onion Services and the clear web has a p-value of < 0.05 (5.29×10^{-17}), thus rejecting the null hypothesis and confirming that Onion Services use different key lengths to the clear web. Onion Services tend to use longer key lengths than on the clear web, increasing security.

Highest TLS version supported	Number of sites	Proportion of sites supporting HTTPS
SSL 2.0	0	0.00%
SSL 3.0	0	0.00%
TLS 1.0	0	0.00%
TLS 1.1	0	0.00%
TLS 1.2	121	61.11%
TLS 1.3	75	37.88%

Table 5: TLS versions supported on Onion Services

Highest TLS version supported	Number of sites	Proportion of sites supporting HTTPS
SSL 2.0	0	0.00%
SSL 3.0	0	0.00%
TLS 1.0	342	11.62%
TLS 1.1	0	0.00%
TLS 1.2	148,207	23.64%
TLS 1.3	478,324	76.30%

Table 6: TLS versions supported on clear web sites

Key Length	Number of sites	Proportion of sites supporting RSA
< 2048	0	0.00%
2048	52	62.62%
3072	0	0.00%
4096	31	37.35%
8192	0	0.00%

Table 7: Key Length of Onion Services

Key Length	Number of sites	Proportion of sites supporting RSA
< 2048	117	0.02%
2048	445,237	90.39%
3072	3,167	0.64%
4096	44,059	8.94%
8192	19	0.00%

Table 8: Key Length of clear web sites

Hypothesis 5 Onion Services are as likely to use Extended Validation certificates as other sites. The proportion of sites having EV certificates are shown in 9. The χ^2 contingent test between Onion Services and other sites has a p-value of < 0.05 (3.98×10^{-33}), thus rejecting the null hypothesis and confirming that Onion Services are more likely to use EV certificates than other sites.

EV certificate	Type of site	Number	Proportion of sites scanned
Yes	Onion Service	18	9.09%
No	Onion Service	180	90.91%
Yes	Clear web	7,069	0.88%
No	Clear web	797,930	99.12%

Table 9: Extended Validation support on different sites

5.1.1 Other data

This section consists of data that has no comparable equivalent in the Crawler.Ninja dataset, but is nonetheless interesting and included in table 10 for reference.

5.2 HTTP Headers

Hypothesis 6 Onion Services are as likely to use Strict Transport Security as other sites. The proportion of sites having an STS header are shown in 11. The χ^2 contingent test between Onion Services and other sites has a p-

Metric	Number	Proportion of sites scanned
Supporting TLS compression	0	0.00%
Supporting TLS 1.3 early data	9	4.55%
Supporting TLS downgrade prevention	196	98.99%
Vulnerable to heartbleed	0	0.00%
Vulnerable to CCS injection	0	0.00%
Vulnerable to client renegotiation DOS	0	0.00%
Vulnerable to ROBOT (weak oracle)	0	0.00%
Vulnerable to ROBOT (weak oracle)	0	0.00%

Table 10: Other metrics concerning Onion Service's TLS setup

value of < 0.05 (5.00×10^{-8}), thus rejecting the null hypothesis and confirming that Onion Services are less likely to use STS than other sites.

STS Header	Type of site	Number	Proportion of sites scanned
Yes	Onion Service	79	12.87%
No	Onion Service	535	87.13%
Yes	Clear web	220,742	22.07%
No	Clear web	779,258	77.93%

Table 11: Strict Transport Security support on different sites

Hypothesis 7 Onion Services are as likely to use a Content Security Policy as other sites. The proportion of sites having a CSP header are shown in 12. The χ^2 contingent test between Onion Services and other sites has a p-value of < 0.05 (6.88×10^{-38}), thus rejecting the null hypothesis and confirming that Onion Services are more likely to use CSP than other sites.

CSP Header	Type of site	Number	Proportion of sites scanned
Yes	Onion Service	160	26.06%
No	Onion Service	454	73.94%
Yes	Clear web	102,296	10.23%
No	Clear web	897,704	89.77%

Table 12: Content Security Policy support on different sites

Hypothesis 8 Onion Services are as likely to use a Permissions Policy as other sites. The proportion of sites having a PP header are shown in 13. The χ^2 contingent test between Onion Services and other sites has a p-value of < 0.05 (0 - less than the smallest floating point number in python), thus rejecting the null hypothesis and confirming that Onion Services are more likely to use PP than other sites.

PP Header	Type of site	Number	Proportion of sites scanned
Yes	Onion Service	116	18.89%
No	Onion Service	498	81.11%
Yes	Clear web	5,623	0.56%
No	Clear web	994,377	99.44%

Table 13: Permissions Policy support on different sites

Hypothesis 9 Onion Services are as likely to use X-Frame-Options as other sites. The proportion of sites having an XFO header are shown in 14. The χ^2 contingent test between Onion Services and other sites has a p-value of < 0.05 (8.44×10^{-42}), thus rejecting the null hypothesis and confirming that Onion Services are more likely to use XFO than other sites.

Hypothesis 10 Onion Services are as likely to use X-Content-Type-Options as other sites. The proportion of sites having an XCTO header are shown in 15. The χ^2 contingent test between Onion Services and other sites has a p-value of < 0.05 (1.61×10^{-123}), thus rejecting the null hypothesis and confirming that Onion Services are more likely to use XCTO than other sites.

XFO Header	Type of site	Number	Proportion of sites scanned
Yes	Onion Service	305	49.67%
No	Onion Service	309	50.33%
Yes	Clear web	256,952	25.70%
No	Clear web	743,048	74.30%

Table 14: X-Frame-Options support on different sites

XCTO Header	Type of site	Number	Proportion of sites scanned
Yes	Onion Service	374	60.91%
No	Onion Service	240	39.09%
Yes	Clear web	215,687	21.57%
No	Clear web	784,313	78.43%

Table 15: X-Content-Type-Options support on different sites

Hypothesis 11 Onion Services are as likely to use a Referrer Policy as other sites. The proportion of sites having an RP header are shown in 16. The χ^2 contingent test between Onion Services and other sites has a p-value of < 0.05 (4.53×10^{-22}), thus rejecting the null hypothesis and confirming that Onion Services are less likely to use an RP than other sites.

RP Header	Type of site	Number	Proportion of sites scanned
Yes	Onion Service	0	0.00%
No	Onion Service	614	100.00%
Yes	Clear web	133,314	13.33%
No	Clear web	866,686	86.67%

Table 16: Referrer Policy support on different sites

5.2.1 Other data

This section consists of data that has no comparable equivalent in the Crawler.Ninja dataset, but is nonetheless interesting and included in table 17 for reference.

Metric	Number	Proportion of sites scanned
Cross Origin Embedder Policy present	46	7.49%
Cross Origin Opener Policy present	50	8.14%
Cross Origin Resource Policy present	52	8.47%
Onion-Location present	111	18.08%

Table 17: Other metrics concerning Onion Service's HTTP setup

6 Discussion

Tor Onion Services are primarily used for hosting HTTP based web sites, but some are used for other purposes, thus the security of how HTTP is configured on these Onion services is paramount to the security of their users.

Onion Services are not configured in any metric in a similar way to sites on the clear web. Onion Services are less likely to support HTTPS, and when they do support it are likely to not support newer versions of the TLS protocol. In spite of this Onion Services do tend to use longer RSA key lengths than sites on the clear web, improving the security of the connection. They're also more likely to use Extended Validation Certificates, providing greater assurances to the service's user about who they're talking to, which is important in cases such as whistle-blowing to news organisations.

Considering application layer security defences in HTTP Onion Services are more likely to use all but two security policies - Strict Transport Security and Referrer Policy. The reason for the complete lack of Referrer Policies on Onion Services is not clear, however the lack of STS can be explained by the general lack of HTTPS on Onion Services which is a prerequisite for enabling STS.

Overall Onion Services do more to protect their users than sites on the clear web, however are lacking in HTTPS deployment. This can be explained by the lack of Certificate Authorities that issue certificates for Onion

Services, and that none offer certificates free of charge¹⁶. More work needs to be done to increase the level of HTTPS deployment on Onion Services to a similar level as on the clear web.

In terms of what could be explored in future work; this research did not look at security vulnerabilities in Onion Services such as Cross Site Scripting attacks, SQL injection, or other such vulnerabilities which are errors of poor coding than of security policy. Additionally it was raised by some (after data collection) that some less knowledgeable users may publish their sites on ports such as 5000, 8080 and 8443. Finally, no tests of the availability of TLS/STARTTLS on SMTP nor FTP were conducted.

A An onion



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¹⁶DigiCert and HARICA are the only CAs to issue such certificates. DigiCert additionally only issues the much more expensive EV certificates. None of the common free CAs such as Let’s Encrypt issue such certificates

¹⁷<https://irl.xyz>

¹⁸<https://www.mpi-inf.mpg.de/>

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